

SHETH L.U.J AND SIR MV COLLEGE

PRACTICAL NO:- 07

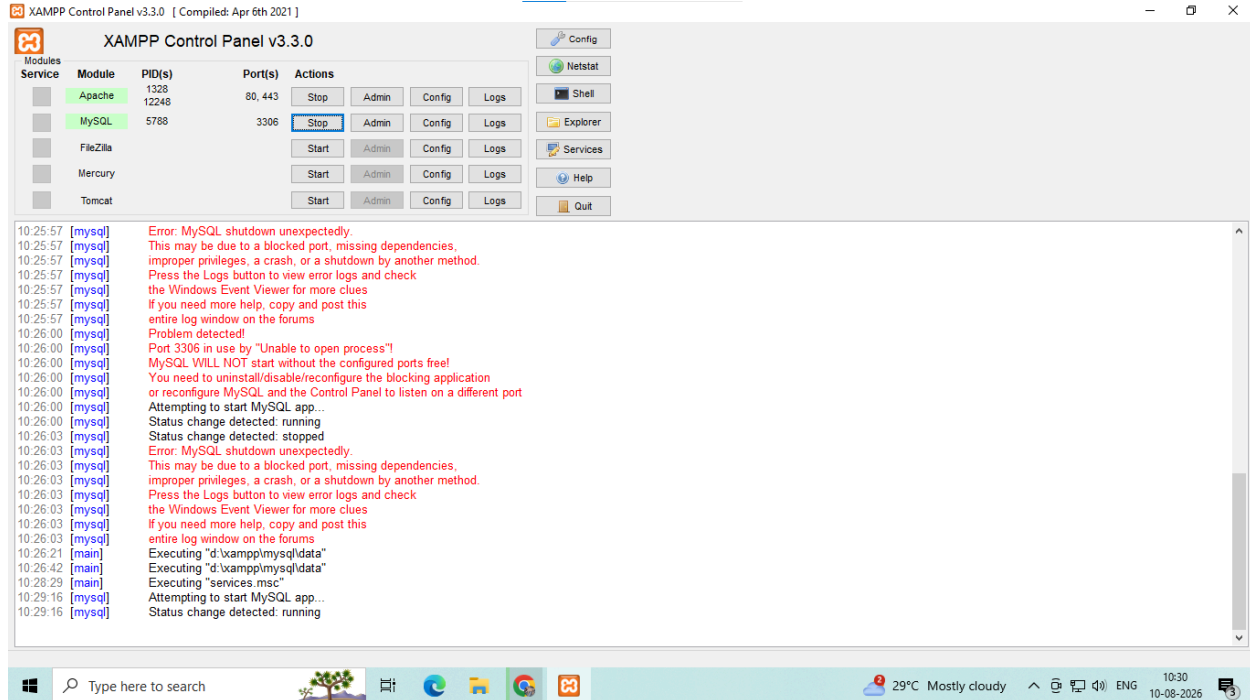
Step 1: Start XAMPP

Open XAMPP Control Panel.

Click Start for:

Apache

MySQL



Step 2: Open phpMyAdmin

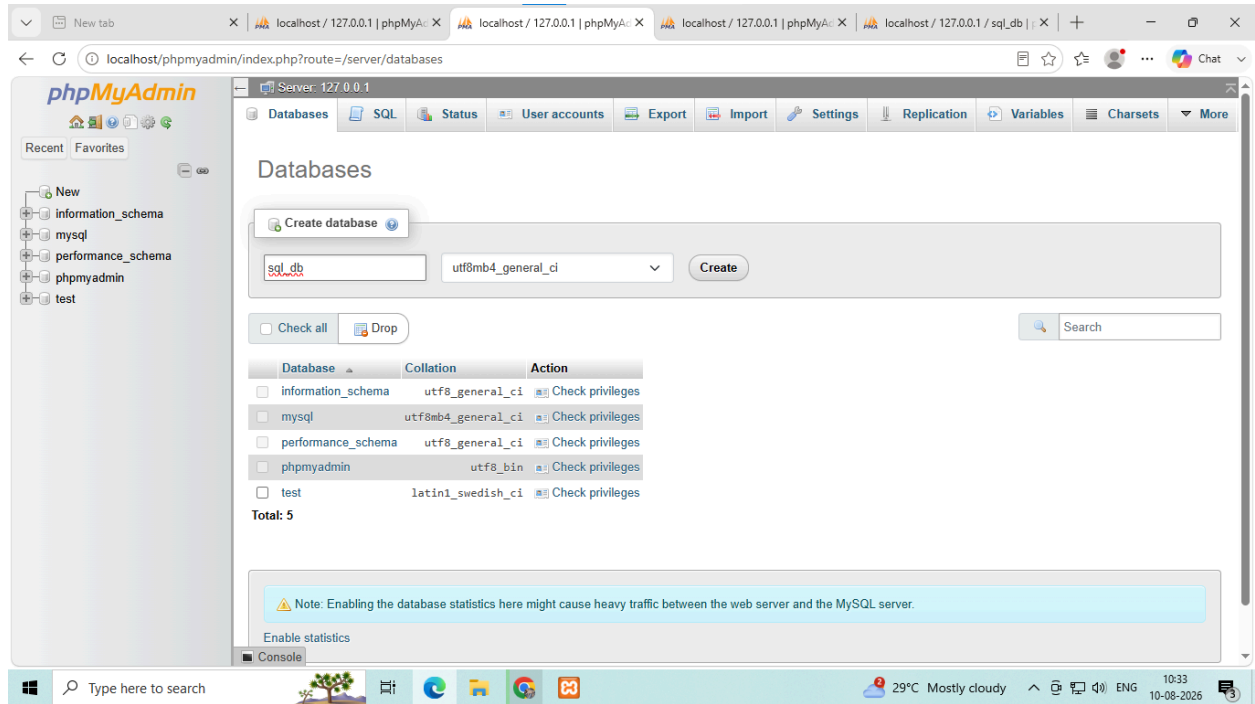
Create a Database

sql_db

Click Create

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Step 3: Download & Setup DVWA

Set Up the SQL Injection Environment

Download DVWA from GitHub:

Go to: <https://github.com/digininja/DVWA>

Click Code → Download ZIP.

- Extract the downloaded ZIP file inside C:\xampp\htdocs\.
- Rename the DVWA-master folder to sql_injection.

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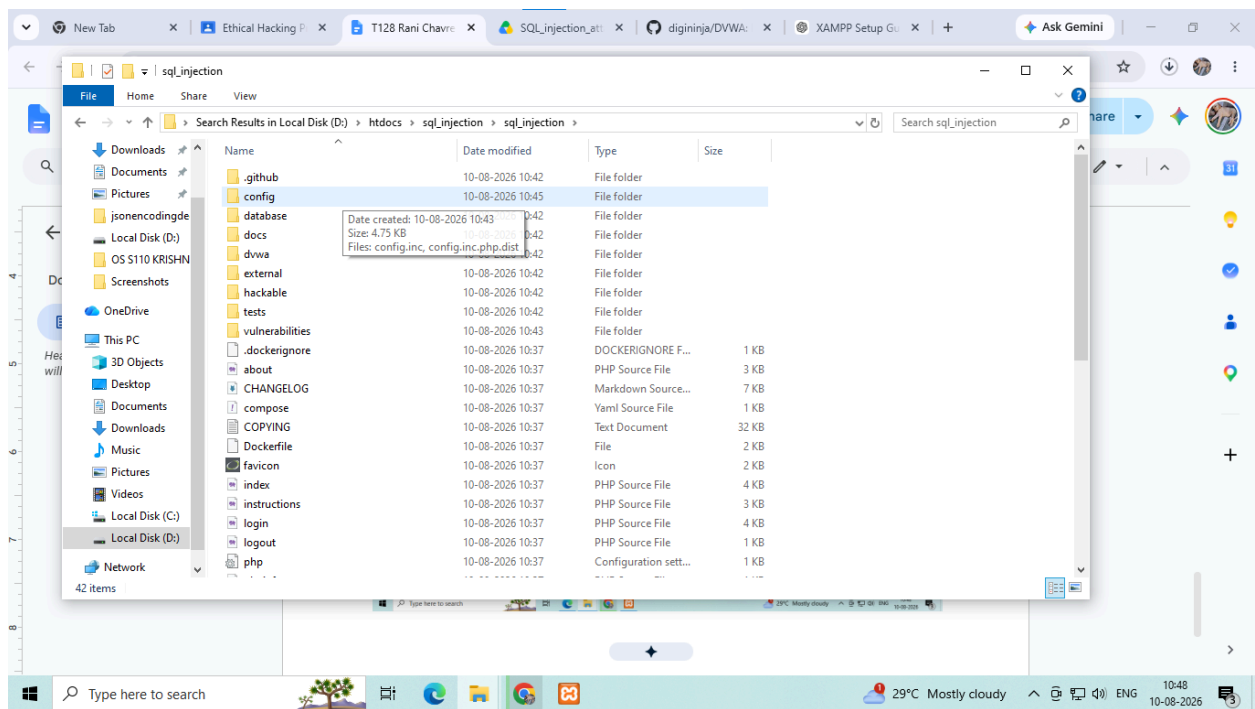
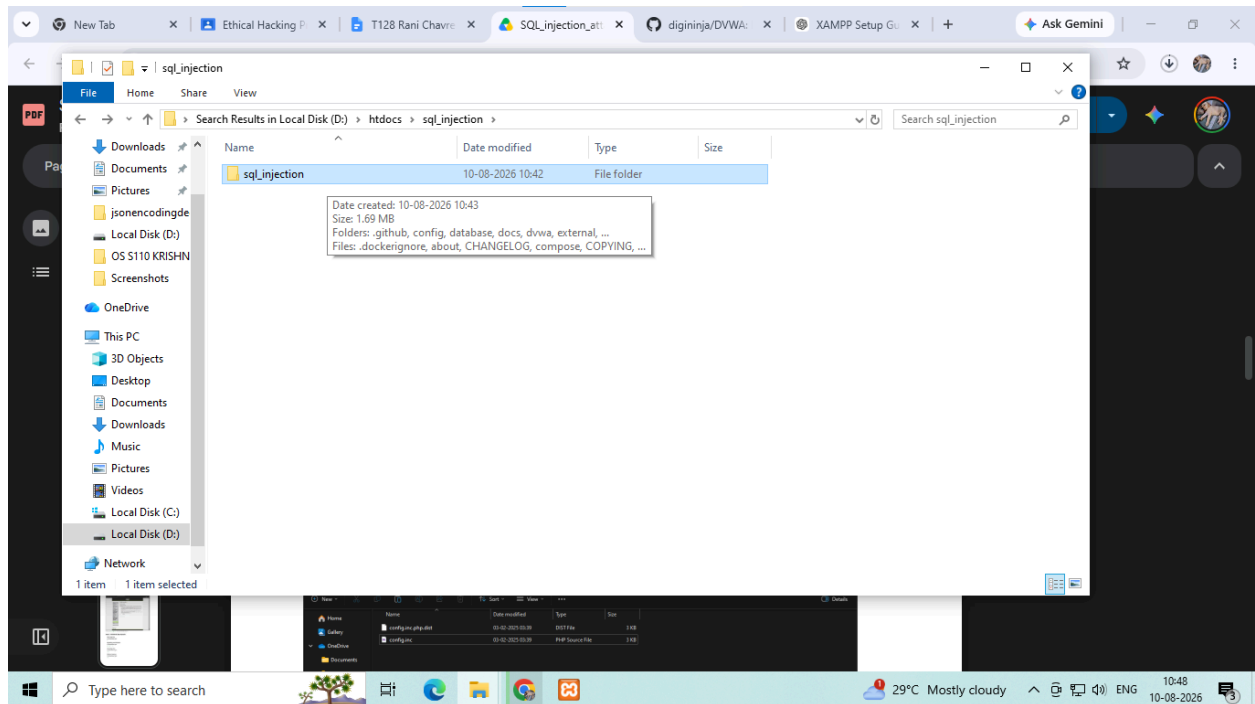
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GitHub repository page for DVWA (Damn Vulnerable Web Application) by digininja. The repository is public and has 316 watches, 5k forks, and 13.5k stars. The repository is public and has 3 branches and 11 tags. The file list includes .github, config, database, docs, dvwa, external/recaptcha, hackable, tests, vulnerabilities, .dockerignore, and .gitattributes. The 'dvwa' directory is highlighted, showing a commit history. The 'About' section describes DVWA as a Damn Vulnerable Web Application (DVWA) with tags for dwwa, hacking, infosec, php, security, sql-injection, and training. It includes a Readme, GPL-3.0 license, Security policy, Activity, 13.5k stars, 316 watching, 5.0k forks, and a Report repository link. The 'Releases' section shows 10 releases, with the latest being 'Vulnerable APIs' from last year.

Windows File Explorer showing the Downloads folder. The folder contains 31 items, including files like DVWA-master, xampp-windows-x64-8.2.12-0-VS16-installer, SQL_injection_attack (1), Unconfirmed 175645.crdownload, T128 Rani Chavre - Practical 4 _ Digital Si..., T128 Rani Chavre - Practical 5_ Key Exch..., and DVWA-master. The 'Last week (19)' section shows a list of files including iris.csv, S110 Krishna Sharma -Practical 7 _ JDBC, xampp-windows-x64-8.2.12-0-VS16-installer, mysql-installer-community-8.0.46.0, S110 Krishna Sharma - Practical 6 _ JSON ..., json-20240303, json-20240303-javadoc (1), mysql-connector-java-5.1.10-bin, json-20240303-javadoc, S110 Krishna Sharma - Practical 8 _ Bread..., S110 Krishna Sharma - Practical 7_ Huffm..., inet-2.1.0.tar.gz, inet-2.1.0, inet-2.1.0-src.tgz, omettp-4.2.2-src-windows, not 6 .docx, VisualStudioSetup (1), and inet-2.1.0.

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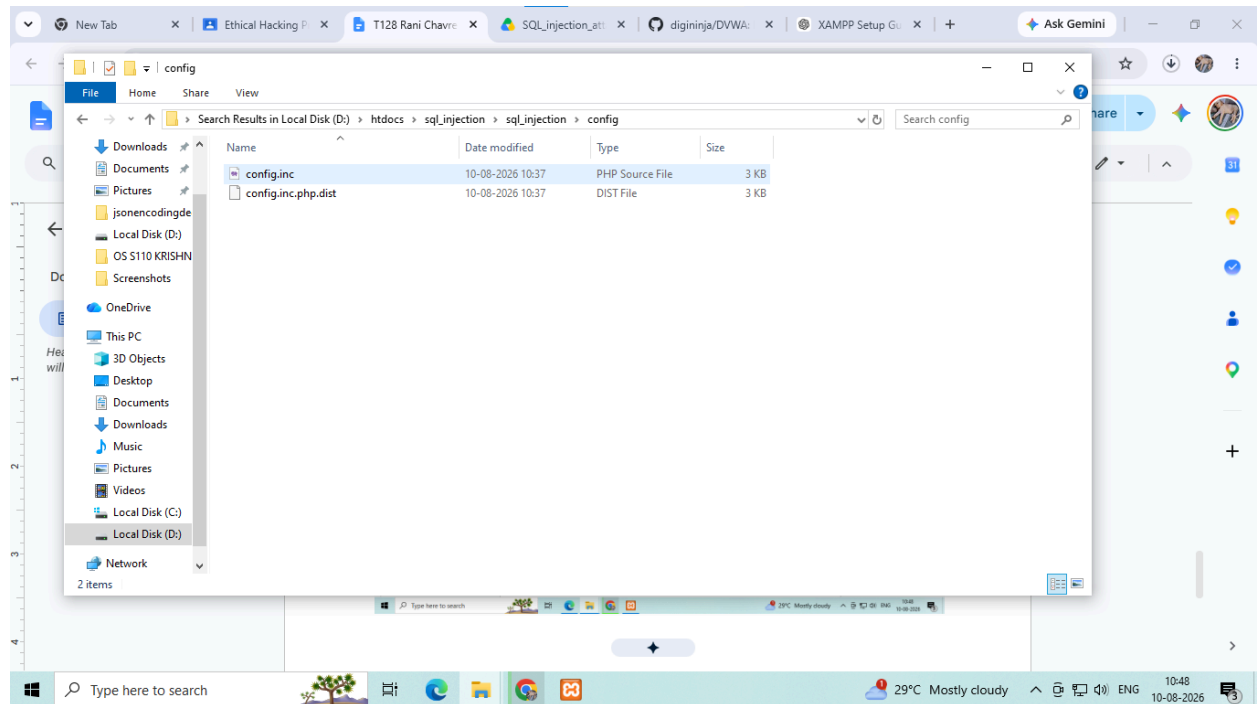


Now go to the sql_injection\config folder

This file should be there config.inc.php.dist copy paste and rename it to config.inc.php

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Modify the Database Credentials

Inside config.inc.php, find this section:

```
$_DVWA[ 'db_user' ] = 'dvw';
```

```
$_DVWA[ 'db_password' ] = 'p@ssw0rd';
```

- ◆ Change it to use MySQL's default root user:

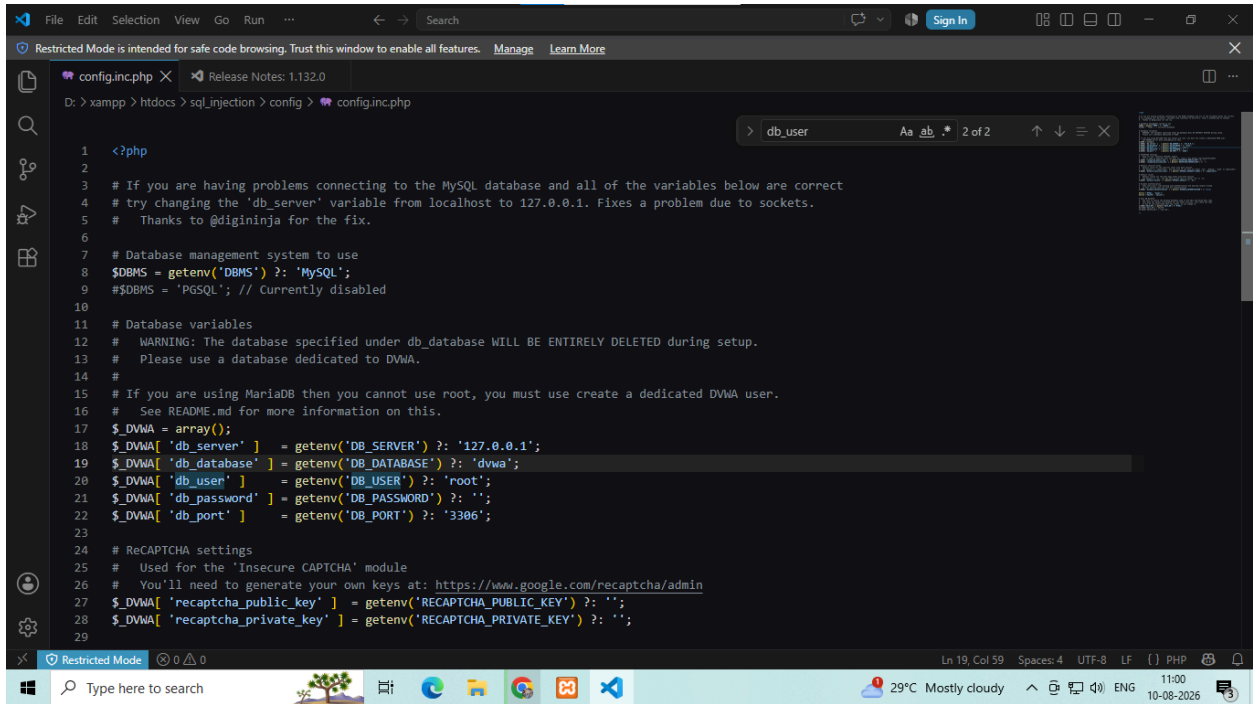
```
$_DVWA[ 'db_user' ] = 'root';
```

```
$_DVWA[ 'db_password' ] = '';
```

Since XAMPP MySQL doesn't have a password for root by default, leaving it empty will Work.

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```
1 <?php
2
3 # If you are having problems connecting to the MySQL database and all of the variables below are correct
4 # try changing the 'db_server' variable from localhost to 127.0.0.1. Fixes a problem due to sockets.
5 # Thanks to @diginiinja for the fix.
6
7 # Database management system to use
8 $DBMS = getenv('DBMS') ? 'MySQL' : 'MySQL';
9 # $DBMS = 'PGSQL'; // Currently disabled
10
11 # Database variables
12 # WARNING: The database specified under db_database WILL BE ENTIRELY DELETED during setup.
13 # Please use a database dedicated to DVWA.
14 #
15 # If you are using MariaDB then you cannot use root, you must use create a dedicated DVWA user.
16 # See README.md for more information on this.
17 $DVWA = array();
18 $DVWA['db_server'] = getenv('DB_SERVER') ? '127.0.0.1' : 'localhost';
19 $DVWA['db_database'] = getenv('DB_DATABASE') ? 'dvwa' : 'dvwa';
20 $DVWA['db_user'] = getenv('DB_USER') ? 'root' : 'root';
21 $DVWA['db_password'] = getenv('DB_PASSWORD') ? '' : '';
22 $DVWA['db_port'] = getenv('DB_PORT') ? '3306' : '3306';
23
24 # ReCAPTCHA settings
25 # Used for the 'Insecure CAPTCHA' module
26 # You'll need to generate your own keys at: https://www.google.com/recaptcha/admin
27 $DVWA['recaptcha_public_key'] = getenv('RECAPTCHA_PUBLIC_KEY') ? '' : '';
28 $DVWA['recaptcha_private_key'] = getenv('RECAPTCHA_PRIVATE_KEY') ? '' : '';
```

Step 4:

Open a web browser.

Type in the URL:

http://localhost/sql_injection/setup.php

Click on Create/Reset Database.



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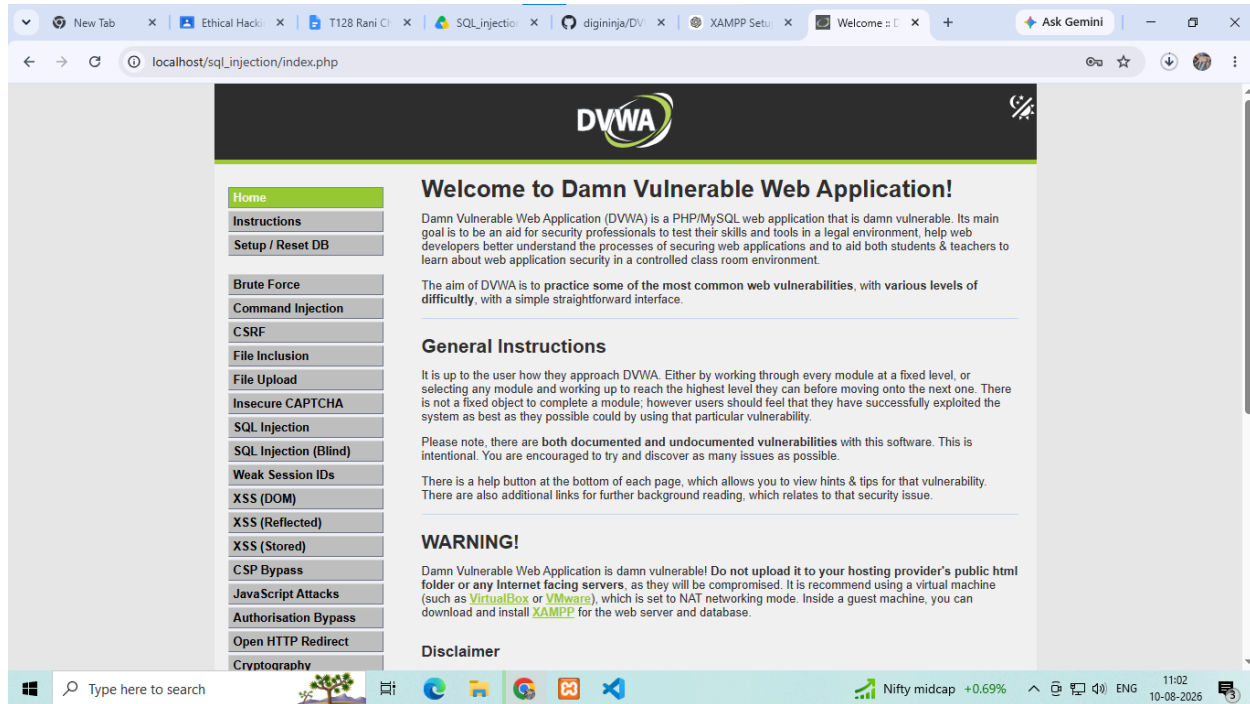
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Step 6: Lower Security Level

Look at the left-side menu.

Click on DVWA Security.

Set the Security Level to Low.



Click on SQL Injection from the left-side menu.

Step 7: Performing SQL Injection

Basic Injection

In the text box, type:

1

Bypassing Authentication

In the text box, type:

a' or '='

Extracting Data

In the text box, type:

1=1

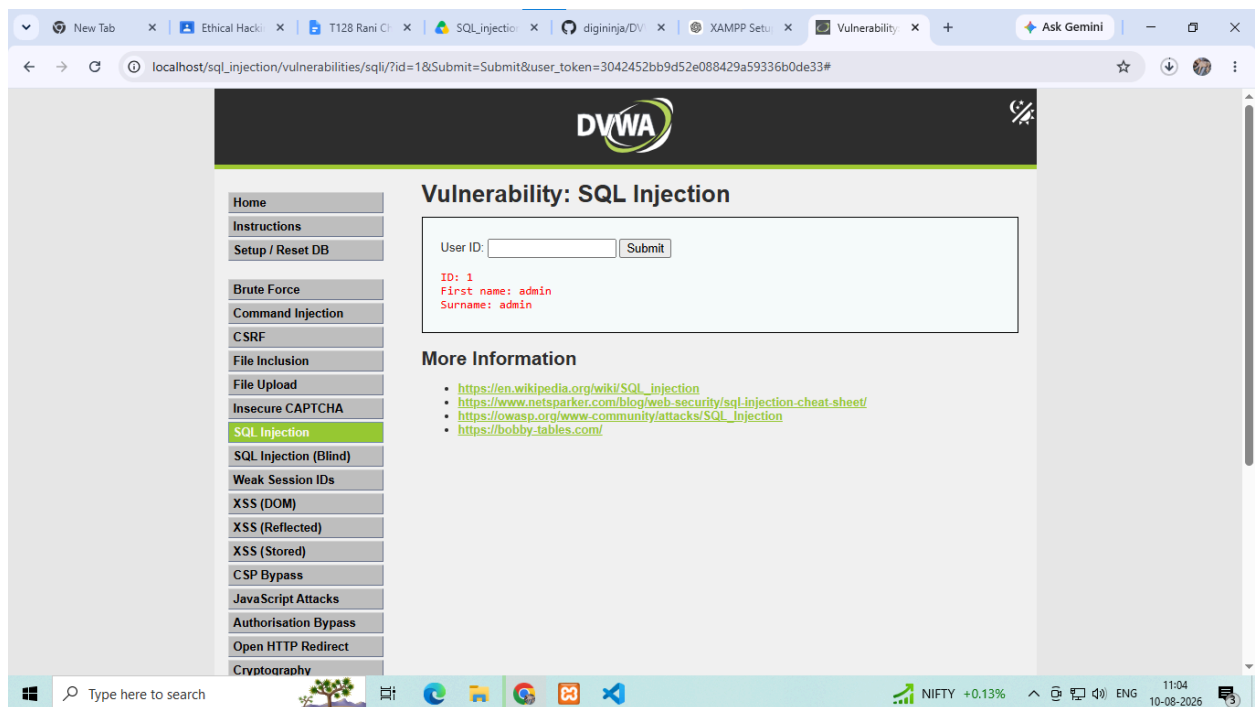
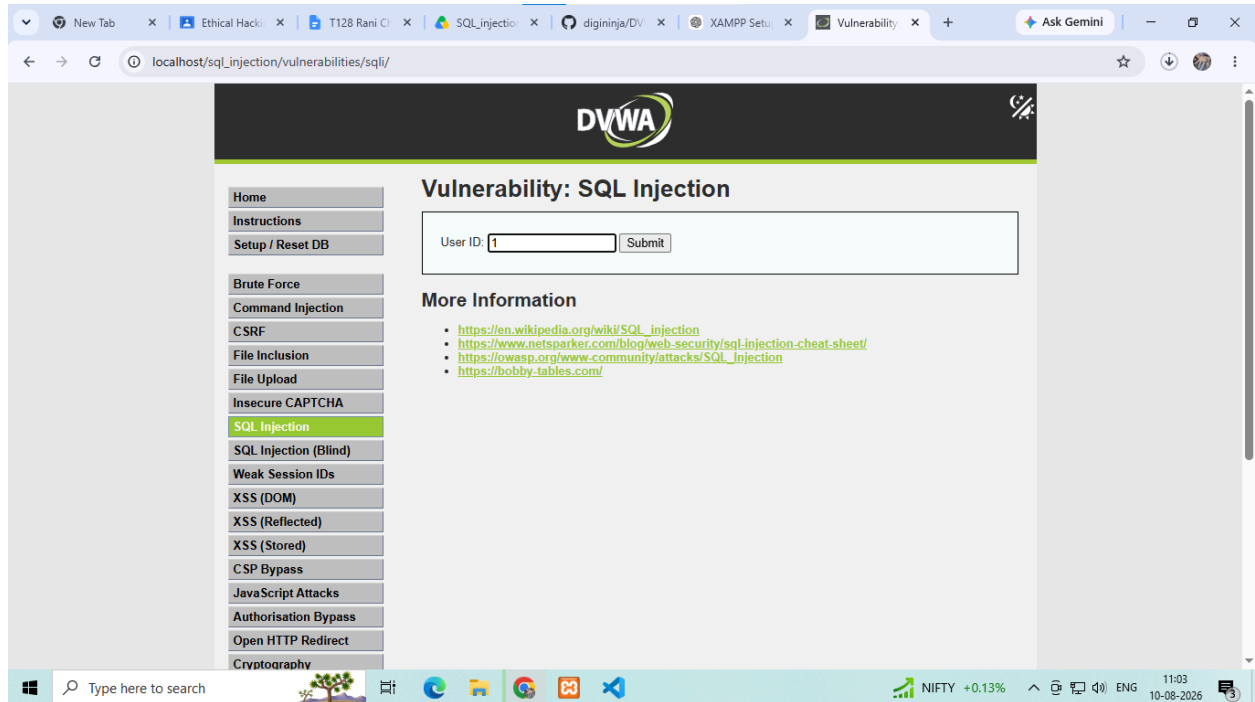
Wildcard Injection

In the text box, type:

1*

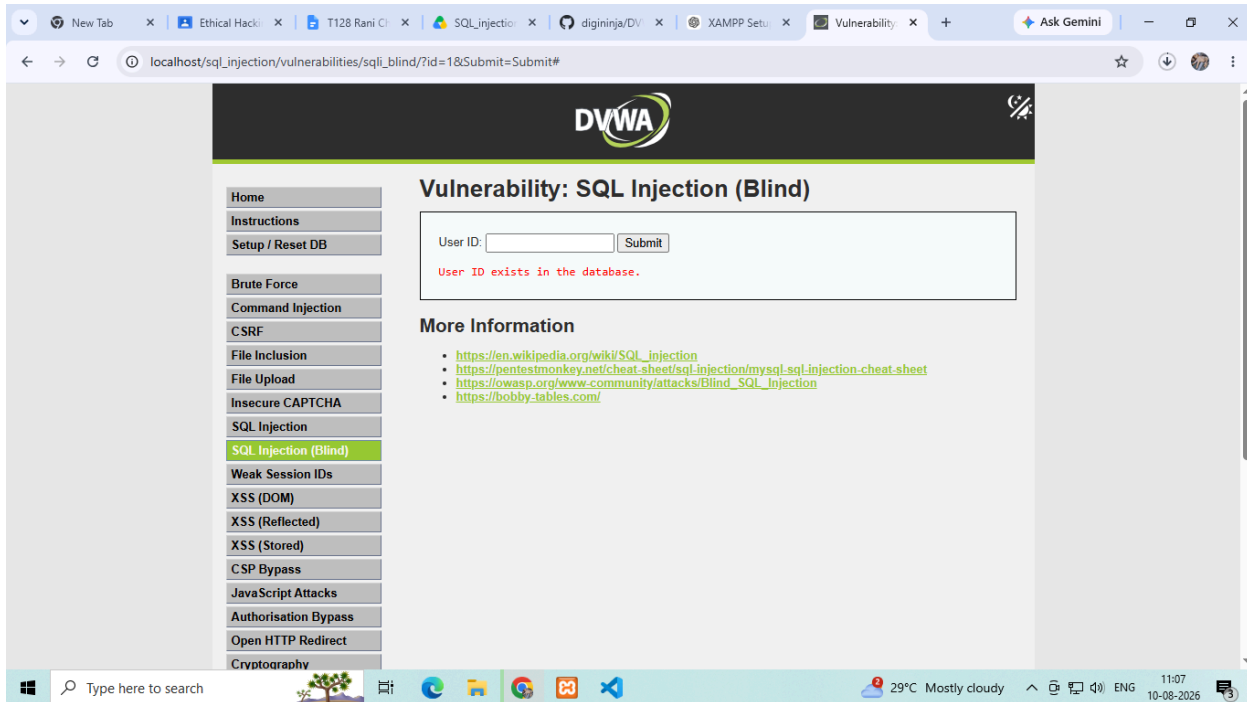
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The screenshot shows the DVWA (Damn Vulnerable Web Application) interface. The browser's address bar displays the URL: `localhost/sql_injection/vulnerabilities/sql_i_blind/?id=1&Submit=Submit#`. The page title is "Vulnerability: SQL Injection (Blind)". On the left, a sidebar menu lists various vulnerabilities, with "SQL Injection (Blind)" selected. The main content area features a "User ID:" input field and a "Submit" button. Below the input field, a red message states: "User ID exists in the database." Under the "More Information" section, several links are provided for further reading on SQL injection.

Home
Instructions
Setup / Reset DB
Brute Force
Command Injection
CSRF
File Inclusion
File Upload
Insecure CAPTCHA
SQL Injection
SQL Injection (Blind)
Weak Session IDs
XSS (DOM)
XSS (Reflected)
XSS (Stored)
CSP Bypass
JavaScript Attacks
Authorisation Bypass
Open HTTP Redirect
Cryptography

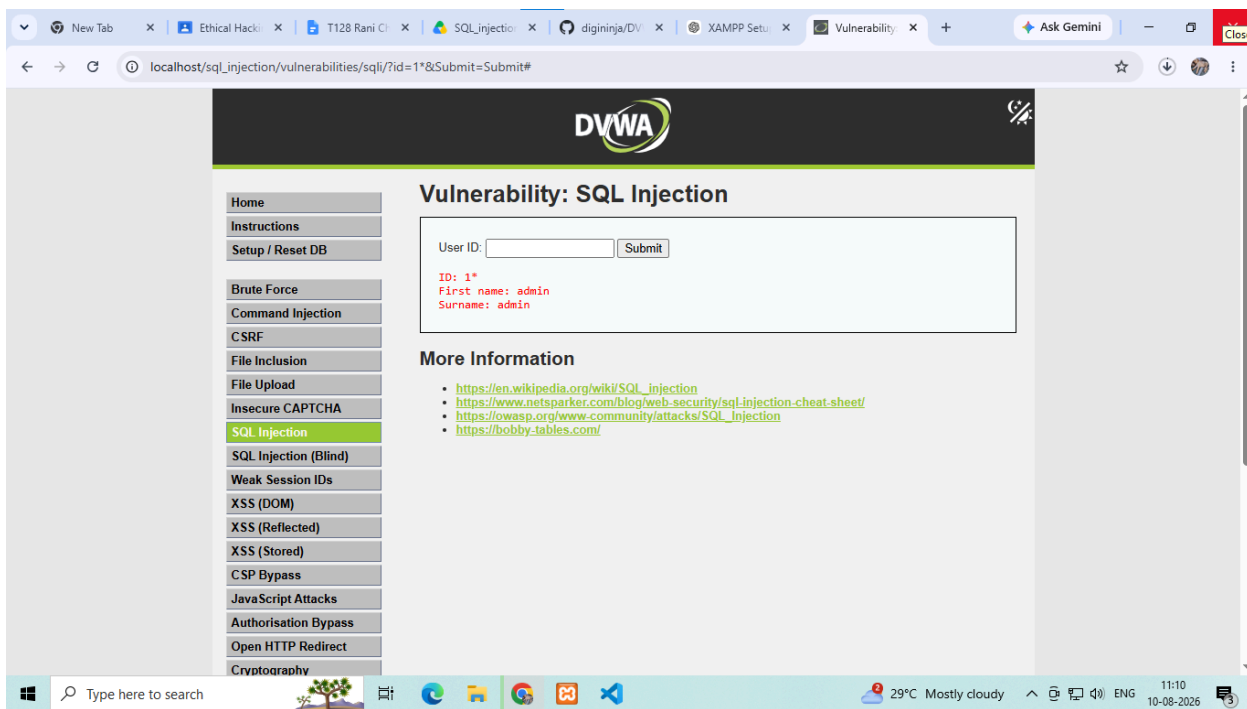
Vulnerability: SQL Injection (Blind)

User ID: Submit

User ID exists in the database.

More Information

- https://en.wikipedia.org/wiki/SQL_injection
- <https://pentestmonkey.net/cheat-sheet/sql-injection/mysql-sql-injection-cheat-sheet>
- https://owasp.org/www-community/attacks/Blind_SQL_injection
- <https://bobby-tables.com/>



The screenshot shows the DVWA interface with the browser's address bar displaying the URL: `localhost/sql_injection/vulnerabilities/sql/?id=1*&Submit=Submit#`. The page title is "Vulnerability: SQL Injection". The sidebar menu is the same as in the first screenshot, but "SQL Injection" is selected instead of "SQL Injection (Blind)". The main content area shows the "User ID:" input field and "Submit" button. Below the input field, a red message displays the results of the query: "ID: 1*", "First name: admin", and "Surname: admin". The "More Information" section contains the same links as the first screenshot.

Vulnerability: SQL Injection

User ID: Submit

ID: 1*
First name: admin
Surname: admin

More Information

- https://en.wikipedia.org/wiki/SQL_injection
- <https://www.netsecracker.com/blog/web-security/sql-injection-cheat-sheet/>
- https://owasp.org/www-community/attacks/SQL_injection
- <https://bobby-tables.com/>

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The screenshot shows the DVWA (Damn Vulnerable Web Application) interface. The browser address bar displays the URL: `localhost/sql_injection/vulnerabilities/sql/?id=1%3D1&Submit=Submit#`. The page title is "Vulnerability: SQL Injection". On the left, a sidebar menu lists various vulnerabilities, with "SQL Injection" highlighted. The main content area shows a "User ID:" input field with the value "1" and a "Submit" button. Below the input field, the output displays: `ID: 1=1`, `First name: admin`, and `Surname: admin`. Under the "More Information" section, there are four links: https://en.wikipedia.org/wiki/SQL_injection, <https://www.netsparker.com/blog/web-security/sql-injection-cheat-sheet/>, https://owasp.org/www-community/attacks/SQL_injection, and <https://bobby-tables.com/>. The Windows taskbar at the bottom shows the system clock as 11:10 on 10-08-2025.

The screenshot shows the DVWA interface with the browser address bar displaying the URL: `localhost/sql_injection/vulnerabilities/sql/?id=a%27+or+%27%27%3D%27&Submit=Submit#`. The page title is "Vulnerability: SQL Injection". The sidebar menu on the left is identical to the first screenshot, with "SQL Injection" highlighted. The main content area shows the "User ID:" input field with the value "a" and a "Submit" button. The output displays three successful login results: `ID: a' or '='`, `First name: admin`, `Surname: admin`; `ID: a' or '='`, `First name: Gordon`, `Surname: Brown`; and `ID: a' or '='`, `First name: Hack`, `Surname: Me`. Below these, it shows `ID: a' or '='`, `First name: Pablo`, `Surname: Picasso`, and `ID: a' or '='`, `First name: Bob`, `Surname: Smith`. The "More Information" section contains the same four links as the first screenshot. The Windows taskbar at the bottom shows the system clock as 11:11 on 10-08-2025.

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The screenshot displays the DVWA web application interface. The browser's address bar shows the URL: `localhost/sql_injection/vulnerabilities/sqli/?id=1&Submit=Submit#`. The DVWA logo is visible at the top. On the left, a sidebar lists various vulnerabilities, with 'SQL Injection' currently selected. The main content area is titled 'Vulnerability: SQL Injection'. It features a form with a 'User ID:' input field and a 'Submit' button. Below the form, the output shows: `ID: 1`, `First name: admin`, and `Surname: admin`. Under the 'More Information' section, there are four links to external resources: https://en.wikipedia.org/wiki/SQL_injection, <https://www.netsparker.com/blog/web-security/sql-injection-cheat-sheet/>, https://owasp.org/www-community/attacks/SQL_injection, and <https://bobby-tables.com/>. The Windows taskbar at the bottom shows the system clock as 11:11 on 10-08-2026, with a temperature of 29°C and weather 'Mostly cloudy'.